nature portfolio

Peer Review File

Globalization of wild capture and farmed aquatic foods

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Reviewers' Comments:

Reviewer #1:
Remarks to the Author:
I think that the paper is well-written, and that the authors have made use of appropriate empirical methods for analyzing the data and presenting results. I recommend the article for publication, pending some minor additions which I think will be informative for the reader and strengthen the empirical points the authors want to make.

I will detail my comments further, below.

Main comments:

1. The authors make the point about the nutritional importance of aquatic foods in key regions of the world, several places in the manuscript. I believe this also to be true, but wanted to see how the results shown in the paper translate into terms that demonstrate food or nutritional security. Without going into details like micro-nutrient intake, can the authors at least show (in macro terms) how much the per-capita consumption values translate into in terms of share of animal protein intake – and how it has changed over time? I think this would be informative and strengthen the argument the authors are trying to make.
2. To strengthen the argument about how the aquatic food system has become globalized, it would be informative for the authors to show how the share of effort in capture exerted in foreign waters has increased over time. This would be particularly true for countries like China, which have deployed more of their fishing fleet in fishing areas farther away from their home waters over time. The authors mention having this data, from the description of production data (starting on line 326). Without breaking into species, even – would this be possible to be shown in terms of the share of total wild catch landed in foreign waters, by major region, over time, given the data that is already at hand?

Reviewer #2:
Remarks to the Author:
This study makes a significant contribution to the resolution of global food system modelling by developing a new database of aquatic food consumption, production and trade, covering a high range of sectors and commodities otherwise aggregated in leading food system models. The sourcing, integration and reconcilaton of data underpinning the database is non-trivial and will help to evaluate the sustainability of decisions concerning consumption, farming and sourcing of aquatic foods. Although greatly impressed by the author's data mining work, I see several areas for development of the manuscript. I have summarised my suggested revisions below. The main improvement which I would like to see addressed in the revised manuscript relate to methodological review which does not adequately discuss the state-of-the art, presentation and analysis of the database at its full sectoral and spatial resolution, and improved data availability.

I hope the authors find these comments constructive towards the improved presentation and dissemination of their work which is highly commendable.

Comments Line(s)/sections
#1 General (Methodology) As the authors suggest, the study makes an important contribution to improving the sectoral resolution and supply chain coverage of aquatic foods in food system models. However, a more thorough review of the state-of-the-art is needed to more explicitly convey the novelty of and need for the model and database presented, including coverage of MRIO models (e.g. FABIO, GLORIA, EXIOBASE, Eora, WIOD etc.), process-based data (e.g. FAO, UN Comtrade, and BACI alone), and CGE/IAM models.
#2 General (Results) Despite the high country and commodity coverage of the database developed and presented, the analysis of aquatic food consumption, production and trade flows within the study remains highly aggregated with mostly regional trade flows reported per food category. This renders the paper less instructive as an empirical analysis and relies on the reader to delve into the database to discover country-level and species-level trends. For example, intra-regional trade analysis concerns the trade flows between 20+ countries in many cases, which have distinct consumption and trade levels, as illustrated in the SI information (Figures 2 and 3). These findings, alongside species-level trade analysis needs more careful attention in the paper, to constitute a significant contribution to food system modelling literature. A cross-cutting network analysis at country and product level is useful towards this end, alongside dedicated figures in the main paper text which elucidate the full database and trends from it, spatially, sectorally and temporally.
#3 Data availability Although the authors share the code to reproduce the ARTIS database, this will likely be prohibitively complex for most readers. In lieu of data in a supplementary file, the database is defacto inaccessible. The contribution of this study to food system modelling scholarship relies on open-access publication of the ARTIS database. I would urge the authors to consider this, via a .txt or .csv file, their website, or Github/Dropbox.
#4 29 It is instructive to comment on which region(s) saw an increase in trade dependence and OECD / non-OECD shares
#5 30 A further comment on what the structure (i.e. network) of aquatic food trade implies for sustainability interventions, e.g. more decentralised than arable or livestock products, highlighting the need for multilateral action
#6 50 risks' without reference to trade or globalisation seems ambigious. Could this be rephrased to make explicit the subject of discussion
#7 62 Further, it seems relevant to comment on the issue of unreconciled bilateral trade data which you help to resolve and which currently prevents traceability of aqautic foods. E.g. the point of dispatch not being the point of production/provenance always.
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think the issue/instance of re-export needs to be carefully explained before summarising the results.
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#14 138 A significant finding which requires more elaboration. Mention of which species are most traded within each aquatic food category would be extremely helpful, alongside a figure/table (in the main text of SI).
#15 Figure 1 Very nice, but maintain sentence case (e.g. % Production exported). Printed, the line shadings are not easily distinguishable. Could these shadings be made more extreme or dashed be used to help the reader interpret and distiguish between trends.
#16 Figure 2a For readers not familiar with net trade flow diagrams, the positive and minus axes could be incorrectly interpreted. Although mentioned in the key, an annotation of the axes (2a) is helpful.
#17 Figure 2b This figure effectively conveys the scale of inter/intra regional trade. Consider adding the region labels on the left axis and values on the right axis, to more closely align with an IO/MFA format
#18 SI Figures (all) Increase font size on figure axis labels, as printed they are illegible
#19 SI Figure 1 Add labels in caption to figure
#20 SI Figure 2 Specify whether this relates to export or re-export
#21 SI Figure 2-3 These are very interesting findings and trends which warrant inclusion in the text, rather than only regional findings
#22 SI Figure 8 and 10 Consider combining or ordering these figures side-by-side (e.g. SI Fig 8 and SI Fig 9) as they are analogous. Combining the figures might be easier, adding the % of fluxes associated with each allocation stage

These comments are also attached in a spreadsheet.

Response to Reviews

Reviewer #1 (Remarks to the Author):

I think that the paper is well-written, and that the authors have made use of appropriate empirical methods for analyzing the data and presenting results. I recommend the article for publication, pending some minor additions which I think will be informative for the reader and strengthen the empirical points the authors want to make.

I will detail my comments further, below.

Thank you for taking the time to review our paper and for your support. We appreciate your comments and provide responses to each below.

Main comments:

1. The authors make the point about the nutritional importance of aquatic foods in key regions of the world, several places in the manuscript. I believe this also to be true, but wanted to see how the results shown in the paper translate into terms that demonstrate food or nutritional security. Without going into details like micro-nutrient intake, can the authors at least show (in macro terms) how much the per-capita consumption values translate into in terms of share of animal protein intake – and how it has changed over time? I think this would be informative and strengthen the argument the authors are trying to make.

Response: We appreciate this comment as improving understanding of the role of aquatic foods in nutrition is one of the primary motivations for developing this database. However, we have not finalized the improved nutrient content database to align with ARTIS to be able to show meaningful patterns between capture and farmed products based on our own analysis. Consequently, the results here would not differ from the share of animal protein intake values from FAO SOFIA, given that our kg per capita values are so similar. Therefore, we referenced FAO and included the suggested statistics in Lines 394-396: "Given the similarity in consumption, this finding reaffirms the nutritional importance of aquatic foods, which supply 17% of animal protein globally, and over 20% of animal protein for over 3.3. billion people."

2. To strengthen the argument about how the aquatic food system has become globalized, it would be informative for the authors to show how the share of effort in capture exerted in foreign waters has increased over time. This would be particularly

true for countries like China, which have deployed more of their fishing fleet in fishing areas farther away from their home waters over time. The authors mention having this data, from the description of production data (starting on line 326). Without breaking into species, even – would this be possible to be shown in terms of the share of total wild catch landed in foreign waters, by major region, over time, given the data that is already at hand?

Response: Unfortunately, FAO production data only includes catch location at a very coarse level (19 marine plus 7 inland FAO Major Fishing Areas - https://www.fao.org/fishery/en/area/search). The disaggregation of the trade data traces trade flows back to the producing nation (generally the flag of the vessel for capture), irrespective of whether that production occurred in the fishing nation's territorial waters. We revised the following sentence to further clarify this (Lines 702-704): "Consequently, our estimates of source country generally represent who harvested or caught the aquatic resource regardless of where it was produced (i.e., distant water fishing would generally be attributed to the flag state)."

We are not currently able to distinguish distant water fishing within trade flows at this time as it requires bringing in a new data set and applying an additional layer of assumptions about how the location of catch relates to trade flows that would need to be interrogated. Essentially, one would need to determine whether a nation catching the same species in two locations exports that species proportionally along all trade routes or is biased toward particular trade routes. Trade and distant water fishing can be structurally related when foreign catch is landed in the host country, sometimes for processing, creating an export from the foreign country to the host country, and possibly back again. This suggests that a simple proportional allocation may not be appropriate. This is a planned next step for the database and we recently secured funding to investigate this issue, but it is not something we can incorporate at this time.

Instead, we do mention the role of distant water fishing in the main text, which now reads (Lines 348-352): "Least developed country net exports are dominated by marine capture products and transfer of aquatic foods from least developed countries are likely even higher when catch by foreign fleets are considered, which represents a related but distinct mechanism by which aquatic resource are removed from countries' waters."

Reviewer #2 (Remarks to the Author):

This study makes a significant contribution to the resolution of global food system modelling by developing a new database of aquatic food consumption, production and trade, covering a high range of sectors and commodities otherwise aggregated in leading food system models. The sourcing, integration and reconcilaton of data underpinning the database is non-trivial and will help to evaluate the sustainability of decisions concerning consumption, farming and sourcing of aquatic foods. Although greatly impressed by the author's data mining work, I see several areas for development of the manuscript. I have summarised my suggested revisions below. The main improvement which I would like to see addressed in the revised manuscript relate to methodological review which does not adequately discuss the state-of-the art, presentation and analysis of the database at its full sectoral and spatial resolution, and improved data availability.

I hope the authors find these comments constructive towards the improved presentation and dissemination of their work which is highly commendable.

Thank you for your support and for your time reviewing our paper to provide these helpful comments. Your deep-dive into the methodology is especially appreciated.

Comments Line(s)/sections
#1 General (Methodology) As the authors suggest, the study makes an important contribution to improving the sectoral resolution and supply chain coverage of aquatic foods in food system models. However, a more thorough review of the state-of-the-art is needed to more explicitly convey the novelty of and need for the model and database presented, including coverage of MRIO models (e.g. FABIO, GLORIA, EXIOBASE, Eora, WIOD etc.), process-based data (e.g. FAO, UN Comtrade, and BACI alone), and CGE/IAM models.

Response: We appreciate this comment as it is important to clearly distinguish our database from other available resources. We added text to address the differences with trade databases (UN Comtrade, BACI and FAO), along with differences with MRIO databases in lines 663-679: "ARTIS also differs from other available trade databases and models in important ways. Databases presenting reported bilateral trade, such as UN Comtrade, do not include source country and are reported in terms of products which generally obscures the species and production method. Closely related or derived databases, such as BACI, thus have the same issues when trying to understand aquatic food origin and species (see below for more details on reported trade data). FAO also provides bilateral trade data on fishery

products, with products at a greater disaggregation than what the 6-digit Harmonized System offers, though this still does not provide sourcing or full species/production method information and the data is currently only available for 2019-2021. Multiregional Input-Output models do resolve flows to the source country and these methods could be adapted to derive a similar database once the species mix of domestically exported products is defined. However, existing input-output databases do not disaggregate aquatic foods at this resolution, generally representing "fisheries" and "food and beverages" as a single or in only a few sectors, as these models are generally constrained by the resolution of national input-output and supply and use tables. A notable exception is FABIO, which disaggregates food and agricultural flows into 130 agriculture, food and forestry products, though seafood is represented as a single group."

#2 General (Results) Despite the high country and commodity coverage of the database developed and presented, the analysis of aquatic food consumption, production and trade flows within the study remains highly aggregated with mostly regional trade flows reported per food category. This renders the paper less instructive as an empirical analysis and relies on the reader to delve into the database to discover country-level and species-level trends. For example, intra-regional trade analysis concerns the trade flows between 20+ countries in many cases, which have distinct consumption and trade levels, as illustrated in the SI information (Figures 2 and 3). These findings, alongside species-level trade analysis needs more careful attention in the paper, to constitute a significant contribution to food system modelling literature. A cross-cutting network analysis at country and product level is useful towards this end, alongside dedicated figures in the main paper text which elucidate the full database and trends from it, spatially, sectorally and temporally.

Response: This paper represents the first presentation of the ARTIS database and its methods. Our goal is for the resulting database to answer many more questions than what can be covered in a single article. We are currently working on other papers that focus in more detail on subsets of the database. For example, I am leading a commentary piece that we hope to time up with this paper that focuses on the flow of products of Russian origin into the US via China. We also hope many others will use the data to answer questions focusing on other species flows and we are developing tools to facilitate this use, as described below in response to #3.

In order to focus the paper's message on the findings of the database as a whole, we had to make a difficult decision about what to include in this first paper. We settled on the focus on marine/inland capture/aquaculture

because, as reported, trade data prevents even this most basic distinction. We therefore show that distinguishing among marine/inland capture/aquaculture sectors is important because they exhibit different global patterns across the various measures of globalization, representing a new and important finding.

However, agree that the original version of the paper did not show the full richness of the dataset. To better illustrate the resolution of the data, we added a new figure (Figure 2), which includes maps containing consumption of foreign products by country and the largest flows by habitat-method group for 2019. As additional panels, we include the top traded species by habitat-method group, colored by source and consuming region. This figure relates to the new text (Lines 293-303): "The patterns of top importers are reflected in the patterns of top consumption of foreign aquatic foods by habitat and production group, though the patterns of top flows are more distinct (Fig 2). For example, although the United States is among the top importers for all four habitat and production groups, the sourcing varies, with the largest marine capture sourcing from Canada and Russia, the largest marine aquaculture sourcing from Chile, and its largest inland aquaculture sourcing from China and Vietnam (Fig 2). The major sourcing flows are closely related to the production geographies of the top traded species by production group (Fig 2). There is very little production method overlap in the top consumed traded species and the distributions are relatively flat, apart from marine aquaculture, which is dominated by white leg shrimp (*Litopenaeus vannamei*) and Atlantic salmon (*Salmo salar*) (Fig 2)." Although these figures are naturally dominated by the largest consumers and species, we feel it strikes the appropriate balance of highlighting some country and species-level patterns, and thereby showing the level of resolution available, without detracting from the broader focus of the paper.

Additionally, we added detail to the country-level results that are presented in the SI (Lines 279-292), which now reads: "At the country level, although some countries rank among the top traders across all production methods, such as China for exports and China, the United States and Japan for imports, many countries are only top traders for one (Fig S2). Between 1996 to 2020, China and Russia took over as the top two marine capture exporters, with approximately half of China's marine capture exports representing re-exported products (Fig S2), which aligns with previous findings that 75% of China's total aquatic food imports are re-exported33. China, the United States, and Japan are the top importers of marine

capture products (Fig S3). Meanwhile, Norway and Chile rank highest in marine aquaculture exports, with the United States and Japan leading imports (Fig S2-3). Inland aquatic food trade is dominated by aquaculture, with the highest exports from Vietnam and China and highest imports by the United States, Japan, and South Korea (Fig S2-3). In general, inland production is oriented more toward domestic consumption and what is exported tends to stay within the region, particularly within Asia (Fig 2; Fig S1).”

#3 Data availability Although the authors share the code to reproduce the ARTIS database, this will likely be prohibitively complex for most readers. In lieu of data in a supplementary file, the database is defacto inaccessible. The contribution of this study to food system modelling scholarship relies on open-access publication of the ARTIS database. I would urge the authors to consider this, via a .txt or .csv file, their website, or Github/Dropbox.

Response: As detailed in the original submission files and the data availability statement, we are making the code, input data files, and the data underlying this paper’s figures and results publicly available in a Zenodo repository. We reserved a doi (10.5281/zenodo.10034319) and the repository will become public along with the paper.

Given the size and complexity of the database, we are also in the process of creating an interactive data visualization and download website. We have already developed the beta version of the API and have hired front end web designers to create the website (funded under NSF #2121238). We are on schedule for the beta version of the website to be ready next summer, ahead of an event to support early career researchers in using the database for their own projects. Finally, we are also developing an R package to facilitate data exploration (https://github.com/Seafood-Globalization-Lab/exploreARTIS). We added a sentence to indicate these resources in the data availability statement (Lines 1263-1265): “Additionally, the full ARTIS database and associated metadata will be available via an API and a website, the development of which is supported by the National Science Foundation (NSF HNDS-I 2121238).”

#4 29 It is instructive to comment on which region(s) saw an increase in trade dependence and OECD / non-OECD shares
Response: We currently show trends in import dependence by region (Fig 3d) and show net trade by region (Fig 2a) and for least developed countries

(Fig S6). While we don't include a figure aggregated by OECD membership, the database indicates country by ISO3 code, facilitating analysis of any particular country grouping of interest.

#5 30 A further comment on what the structure (i.e. network) of aquatic food trade implies for sustainability interventions, e.g. more decentralised than arable or livestock products, highlighting the need for multilateral action
Response: We appreciate this comment and added text later in the manuscript with a citation to support the sustainability issues related to distancing consumers from production, noting that the ultimate impacts will depend on the management context. The text now precedes the previous text focused on the problems of complex supply chains for traceability. The text now reads (lines 329-336): "Increasing global trade, along with distant water fishing, drive an expanding divide between aquatic food production and consumption[36]. Distancing consumers from the environmental impacts stemming from production has been proposed to reduce incentives for improving environmental standards and weaken price signals related to fishery sustainability[37]. However, the specific impacts resulting from trade will depend upon the environmental management context. Complex international supply chains also pose a challenge for traceability, raising sustainability concerns, including risk of mislabeled[38] and illegally sourced[39] products entering markets."

#6 50 risks' without reference to trade or globalisation seems ambigious. Could this be rephrased to make explicit the subject of discussion
Response: We modified the sentence to now read "However, risks stemming from globalization include accelerating the nutrition transition to unhealthy diets12, undermining domestic production by suppressing prices13, exposing local markets to international shocks[14], degrading local environments to meet distant market demand[15,16], and facilitating shifting production to locations with relaxed environmental and labor regulations[17,18]." (Lines 69-74).

#7 62 Further, it seems relevant to comment on the issue of unreconciled bilateral trade data which you help to resolve and which currently prevents traceability of aqautic foods. E.g. the point of dispatch not being the point of production/provenance always.
Response: We added to this sentence to indicate the issue of trade data not indicating country of harvest, in addition to discussing this issue in further detail in the methods. The text now reads: "Unfortunately, our understanding of the distribution of global benefits and risks is limited by low species resolution of global trade data relative to the vast diversity of

aquatic foods, and because internationally reported trade data does not indicate the country of harvest." (Lines 76-79).

#8 77 Insert a webpage link to the database
Response: We address this point in more detail in response to point #3, but briefly, we have archived the code and key output files in a Zenodo repository with a reserved DOI (10.5281/zenodo.10034319). This will become public upon publication. We have also built an API and hired a company that is designing a website to make it easier to explore the database.

#9 96 Between 2 and 2.1, it would be helpful to introduce each section of the results and summarise the content as an aid to the reader
Response: As the last paragraph of the introduction provides an overview of the three results sections, we feel this addition would sound repetitive.

#10 100 Could this be contextualised to non-aquatic agricultural trade to indicate whether a similar trend is observed
Response: We were not able to find a reference with an appropriately comparable statistic, so we estimated global exports for cereals over the same period using FAOSTAT, which grew 2.2-fold. However, we are not including the specific 2.2 statistic in the paper because the FAO changed its methods for the Food Balance Sheets in 2014 and we are not able to separate food and non-food trade, so we are not able to make an exact comparison over the study period. Instead, we edited the sentence to read "Aquatic food exports more than doubled from 1996-2019 (28.1 to 59.2 mil t; Fig 1a), on par with cereal products[22]." (Lines 132-133).

#11 110-111 This point warrants careful explanation, referring to re-export of imported products to reach (mostly) land-linked countries, particularly given the large re-export share in overall trade.
Response: We added text to clarify this point (Lines 143-148): "Trade statistics do not generally distinguish between exports source from domestic production and those sourced from imported products, including products moving through intermediate countries or undergoing foreign processing. Consequently, total exports represent an even higher percentage of production. We estimate total exports reached 33.5% of production in 2019, slightly lower than the FAO estimates 36% for all aquatic resources[1]."

#12 112 Is this a false equivalence if it refers to export of production and not re-export? As above, I think the issue/instance of re-export needs to be carefully explained before summarising the results.
Response: While the original sources are not entirely clear, it is our best understanding that their statistic is directly comparable to the statistic which includes exports sourced from foreign/imported products. This is because global statistics do not historically differentiate between products of domestic and foreign origin. We revised the text mentioned in the previous response to clarify this point.

#13 122-127 This sections seems to interrupt the flow and might be better placed in the results introduction.
Response: We moved the text up to the beginning of the section, now falling immediately after the first sentence of the section (Lines 126-131).

#14 138 A significant finding which requires more elaboration. Mention of which species are most traded within each aquatic food category would be extremely helpful, alongside a figure/table (in the main text of SI).
Response: We now include a figure indicating the top species consumed that are of foreign sourcing for each production method group, along with fill color indicating the sourcing and consumption region (Figure 2).

#15 Figure 1 Very nice, but maintain sentence case (e.g. % Production exported). Printed, the line shadings are not easily distinguishable. Could these shadings be made more extreme or dashed be used to help the reader interpret and distiguish between trends.
Response: We corrected the axis labels to maintain sentence case across all figures. We also made the lines thicker and increased the transparency of the shaded bands on panel c.

#16 Figure 2a For readers not familiar with net trade flow diagrams, the positive and minus axes could be incorrectly interpreted. Although mentioned in the key, an annotation of the axes (2a) is helpful.
Response: We added "net import" and "net export" with arrows on the right-hand side of the panel to help communicate the correct interpretation of the positive and negative values.

#17 Figure 2b This figure effectively conveys the scale of inter/intra regional trade. Consider adding the region labels on the left axis and values on the right axis, to more closely align with an IO/MFA format

Response: We updated the figure to have exporting regions labeled on the left, importing regions labeled along the top and the trade volume on the right.

#18 SI Figures (all) Increase font size on figure axis labels, as printed they are illegible
Response: We updated the font sizes across all figures in the main text and SI to ensure better readability.

#19 SI Figure 1 Add labels in caption to figure
Response: We added titles to each of the subplots to indicate habitat/production method. We also adjusted the margins to reduce the white space.

#20 SI Figure 2 Specify whether this relates to export or re-export
Response: We updated the figure to clarify this point and to add more information. The figure now has the bars ranked by total exports (as before), but we now fill the stacked bar graph by the export source (domestic versus foreign versus error).

#21 SI Figure 2-3 These are very interesting findings and trends which warrant inclusion in the text, rather than only regional findings
Response: We expanded the text on the original submission lines 171-180 to discuss some additional country level findings. We also added another layer of information to SI Figures 2 and 3 (as indicated in response to #20). In addition to this, we created a new figure (Figure 2) to show additional country and species level resolution.

#22 SI Figure 8 and 10 Consider combining or ordering these figures side-by-side (e.g. SI Fig 8 and SI Fig 9) as they are analogous. Combining the figures might be easier, adding the % of fluxes associated with each allocation stage
Response: We added the percentages onto SI Figure 8 and removed the previous SI Figure 10.

Reviewers' Comments:

Reviewer #1:
None

Reviewer #2:
Remarks to the Author:
I am pleased by the extensive revisions undertaken by the authors in response to my comments and satisfied all of my suggestions were comprehensively addressed in the revised manuscript. I thank the authors for their further methodological documentation of the modelling procedures used and work to improve the accessibility of figures and data. The final manuscript represents an impressive contribution to the assessment of global marine food system flows and I am delighted further applications of the dataset are planned.